

Rubhav Nayak

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Objective

MSc Electrical Engineering student at TU Delft (Signals & Systems) with hands-on experience in real-time FPGA-based systems, audio signal processing, and graduate-level coursework in sensor signal processing, estimation, detection, distributed systems and machine learning. Seeking an internship in signal processing or embedded DSP within the Netherlands, where I can contribute my skillsets while developing industry experience.

Education

MSc, Electrical Engineering (Signals and Systems) | Delft University of Technology **September 2024 – July 2027**

- **Key Courses:** Statistical Digital Signal Processing | Machine Learning | Control System Design | Applied Convex Optimization | Estimation and Detection | Probabilistic Sensor Fusion | Data Compression | Advanced Digital Image Processing | Sensor Signal and Data Processing | Distributed Signal Processing | Wavefield and MRI Imaging

BSc, Electrical Engineering (with Honors) | University of Illinois at Urbana-Champaign **August 2019 – May 2023**

- **GPA:** 3.66/4.0
- **Key Courses:** Computer Systems and Programming | Data Structures | Analog Signal Processing | Digital Signal Processing | Fields and Waves | Digital Systems Laboratory | Electronic Circuits | Sensors and Instrumentation | Power Circuits and Electromechanics | Semiconductor Devices | Image and Video Processing | Control Systems

Projects

Image Restoration | Course Project, EE4C03 Statistical DSP, TU Delft **October 2025**

- Implemented a custom motion blur filter and Wiener filter from scratch in MATLAB; custom blur matched MATLAB's `fspecial()` benchmark with SSIM difference < 0.005
- Derived the MMSE-optimal Wiener solution estimating signal power spectrum from the distorted image alone; achieved SSIM of 0.40 vs. 0.06 for inverse filtering under Gaussian noise ($\sigma^2 = 10^{-3}$)

Edge-Aware Image Editing | Course Project, AP3132 Advanced Digital Image Processing, TU Delft **June 2025**

- Implemented Local Laplacian Filtering from scratch in MATLAB, building Gaussian/Laplacian pyramid decomposition and a nonlinear remapping function that distinguishes fine-scale detail from strong edges, enabling edge-aware detail enhancement, image smoothing, and HDR tone mapping
- Benchmarked against MATLAB's built-in `locallapfilt()` and a bilateral filter; demonstrated comparable edge-preservation quality with full parametric control over detail (α) and edge contrast (β)

Distributed Consensus Algorithms | Course Project, EE4540 Distributed Signal Processing, TU Delft **June 2025**

- Implemented Randomized Gossip and PDMM for distributed average and median consensus over a 201-node sensor network in Python, achieving faster convergence with PDMM
- Analyzed robustness under simulated sensor and transmission failures, comparing average and median consensus performances.

Real-Time Object Tracker | Course Project, ECE437 Sensors and Instrumentation, UIUC **December 2022**

- Implemented real-time light-source tracking at 20 FPS on a Xilinx FPGA system, using OpenCV for brightness-based detection and a motor-driven correction loop to keep the source centered in frame
- Wrote I2C, SPI, and GPIO finite state machines in Verilog for sensor communication across a CVM300 imaging sensor, LSM303 IMU, and H-bridge motor controller
- Built a multithreaded Python pipeline for high-speed FPGA-to-PC image transfer using BTPipeOut FIFOs, achieving 30 FPS data throughput independent of IMU processing

EchoRanger | Course Project, ECE420 Embedded Digital Signal Processing, UIUC **April 2022**

- Implemented a Java-based Android SONAR ranging application based on Graham's chirp-correlation method, emitting a 10ms chirp and computing cross-correlation between the transmitted and received signals to estimate distance
- Achieved $\pm 0.2\text{m}$ accuracy at 3m range; validated by comparing Android output against a Python reference implementation on identical audio recordings

Skills

Software Skills: Python (NumPy, SciPy, scikit-learn, OpenCV) | MATLAB (Simulink, Image Processing Toolbox) | C | C++ | Java | Verilog | Git

Languages Spoken: English (Native), Hindi (Native), Odia (Native), French (B2), Dutch (A2)

Experience

Teaching Assistant | University of Illinois | Champaign, IL, USA

Jan 2022 – May 2023

- Evaluated lab assignments and supported ~30 students per semester in the Digital Signal Processing Laboratory course

Member | IEEE-HKN Honor Society – Alpha Chapter | Champaign, IL, USA

Jan 2021 – May 2022

- Conducted exam review sessions and hosted office hours for students in 100 and 200-level electrical engineering classes
- Mentored students regarding prospective classes they could take or clubs they could join based on their current and broader interests
- Contributed to the extensive database of resources and guides maintained by the chapter